

Panithan Lertsuntivit

Panithan@berkeley.edu | [linkedin.com/in/panithan-lertsuntivit/](https://www.linkedin.com/in/panithan-lertsuntivit/) | panithan-lertsuntivit.github.io

EDUCATION

University of California, Berkeley

B.S. Mechanical Engineering

Expected December 2025

GPA: 3.73

Coursework: Mechatronic Design, Introduction to Flight Mechanics, Fluid Mechanics, Dynamic Systems and Feedback, Embedded Microprocessor Based Systems, Dynamics, Thermodynamics, Introduction to Solid Mechanics

Programs / Languages: SolidWorks, AutoCAD, Onshape, ESP-IDF, LabVIEW, Python, MATLAB, C

PROFESSIONAL EXPERIENCE

UC Berkeley Mechanical Engineering

Teaching Assistant

August 2024 - Present

Berkeley, CA

- Collaborate regularly with the professor and other teaching assistants to ensure scheduled course progression
- Mentor over 300 students across ME 103 [Experimentation and Measurements] and ME 100 [Electronics for IOT] by hosting weekly discussions, clarifying topics in office hours, and designing enhancements for student assignments
- Guide weekly lab sections with hands-on experience using microcontrollers, electronic sensors and actuators

National Tsing Hua University

Yang Lab for Biomechanics | *Research Intern*

June 2025 - August 2025

Hsinchu, Taiwan

- Rapidly learned the principles of biomechanics for immediate contributions to active research through prototyping
- Iteratively designed and validated a fluid mixing adapter to produce a stable mixture comprised of ink and mucus
- Modeled and coded a controllable fluid release system to build a framework for iterative ink pattern testing

ORGANIZATIONS

Aerospace SAE at Berkeley

Director of Avionics

September 2024 - Present

Berkeley, CA

- Engineered and validated a thrust test rig for propeller and motor tests in static and dynamic wind conditions
- Implemented Molex connectors and integration geometries for secure electrical and physical connections; facilitating a sub-2-minute fully functional wing panel assembly on the flight line for the 2025 SAE Aero Design Competition

PROJECTS & INTERESTS

eV/STOL

August 2025 - Present

- Engineer an aircraft capable of vertical / short take off and landing for increased maneuverability and efficiency
- Integrate electrical components to support a maximum take off weight of 18 N with a cruising speed of 13 m/s

Real-Time Plotter

January 2024 - May 2024

- Developed a wireless ESP32-based plotter using Espressif's IoT Development Framework (ESP-IDF), capable of drawing geometric shapes at a frequency of 65 Hz directed from a LabVIEW graphical user interface
- Optimized system efficiency through multitasking between real-time motor actuation and wireless communication

Mobile Desk

September 2023 - December 2023

- Designed and created a portable desk that conformed to GD&T standards for fits between part for integration
- Manufactured aluminum components using manual lathes and mills in the UC Berkeley Etcheverry Machine Shop

Wall Climbing Vehicle

January 2023 - May 2023

- Designed a remote control vehicle capable of maneuvering on a vertical surface using SolidWorks and Onshape
- Prototyped various wooden living hinges, achieving rapid production of flexible laser-cut wood geometries

Fields of Interests: Robotics, Mechatronics, Mechanical Design

Hobbies: Exploring Cultural Landmarks and Establishments and Asian Gourmet Cooking